BIOL-8 Exam 01 — Answer Key

Modules 01-06

Part A: Multiple Choice (30 points)

Q#	Answer	Explanation
1	C	Correct: Not all organisms use photosynthesis (only autotrophs).
2	B	Correct: Fertilizer amount is the variable manipulated by the experimenter.
3	A	Correct: Organelle → Cell → Tissue → Organ → Organ System.
4	B	Correct: An ecosystem includes the community of organisms plus abiotic factors.
5	B	Correct: Natural selection favors traits that improve survival/reproduction.
6	C	Correct: Atomic number equals the number of protons (6).
7	A	Correct: Covalent bonds form by sharing electron pairs.
8	C	Correct: Water has *high* cohesion (sticks to itself), not low cohesion.
9	C	Correct: pH < 7 is acidic; pH 2 is strongly acidic.
10	B	Correct: Carbon can form 4 stable covalent bonds.
11	C	Correct: Dehydration synthesis removes water to join monomers.
12	D	Correct: Carbohydrates provide the fastest cellular energy.
13	A	Correct: Peptide bonds link amino acids.
14	C	Correct: Saturated fats have max hydrogens and no double bonds.
15	B	Correct: Denaturation is the loss of functional shape.
16	B	Correct: "All living things are made of cells" is a cell theory principle.
17	D	Correct: Eukaryotes have a membrane-bound nucleus; prokaryotes do not.
18	B	Correct: Golgi apparatus processes and packages proteins.

Q#	Answer	Explanation
19	A	Correct: Mitochondria have their own circular DNA/ribosomes.
20	C	Correct: Cell walls are found in plants but not animals.
21	B	Correct: Fluid mosaic = fluid bilayer + embedded proteins.
22	C	Correct: Facilitated diffusion requires transport proteins (no energy).
23	A	Correct: Water moves into the cell (higher solute) → cell swells.
24	A	Correct: Na^+ / K^+ pump moves ions *against* gradients using ATP.
25	B	Correct: Glycoproteins act as ID tags for recognition.
26	B	Correct: ATP is the cell's energy currency.
27	B	Correct: Enzymes lower activation energy to speed up reactions.
28	B	Correct: Respiration breaks down glucose to release energy as ATP.
29	C	Correct: Oxygen is the final electron acceptor.
30	C	Correct: Anabolism builds larger molecules from smaller ones (requires energy).

Part B: Fill in the Blank (11 points)

Q#	Answer	Acceptable Alternatives
31	homeostasis	—
32	ionic	—
33	hydrolysis	—
34	tertiary	—
35	rough	rough ER
36	lysosomes	—
37	osmosis	—
38	isotonic	—
39	induced fit	induced-fit
40	fermentation	anaerobic respiration
41	nucleus	—

Part C: Free Response Rubric (9 points)

Students choose 3 of 5. Each response scored 3 points.

Scoring Guide per Question

Points	Criteria
3	Complete, accurate, well-organized response with specific details
2	Mostly accurate with minor gaps or limited examples
1	Partial understanding; significant information missing
0	Incorrect, off-topic, or blank

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1. Water and Life

Key points (any 3 properties + example):

- **Cohesion/Adhesion**: Water sticks to itself/surfaces → plant transport.
- **Specific Heat**: Resists temperature change → stable environment/body temp.
- **Solvent**: Dissolves many things → nutrient transport in blood.
- **Ice Density**: Ice floats → insulates lakes in winter.

2. The Protein Path

Expected sequence:

1. **Ribosome**: Makes the protein.
2. **Rough ER**: Folds/modifies the protein.
3. **Golgi Apparatus**: Packages/sorts the protein.
4. **Vesicle**: Transports it to edge.
5. **Cell Membrane**: Releases it (secretion).

3. Moving Across Membranes

Passive Transport:

- No energy needed. Moves *down* gradient.
- Example: Oxygen diffusion, Osmosis.

Active Transport:

- Energy (ATP) needed. Moves *against* gradient.
- Example: Sodium-potassium pump.

4. Respiration vs. Fermentation

Aerobic Respiration:

- Needs Oxygen? Yes.
- ATP Yield? High (~30-32).
- Location: Mitochondria + Cytoplasm.

Fermentation:

- Needs Oxygen? No.
- ATP Yield? Low (2).
- Location: Cytoplasm only.

5. Macromolecule Structure

Choose 2:

- **Carbohydrate**: Monosaccharide → Energy.
- **Lipid**: Fatty Acid → Storage/Membrane.
- **Protein**: Amino Acid → Enzyme/Structure.
- **Nucleic Acid**: Nucleotide → Genetics.

End of Answer Key